S.S.M Game Engine Documentation

This document explains how our 2-player platformer engine works and what each part of the code does.

1. What This Engine Is

This is our small 2-player platformer engine.

It's inspired by Ultimate Chicken Horse, but way simpler.

You can build the level first, then switch to play mode and try to beat it.

The idea is that the engine should be reusable later when we make an actual game.

The engine isn't "finished," it's just the base we're building on.

2. How the Game Works (basic loop)

Start Screen

The game starts on a plain grey screen that says "Start."

You click anywhere, and it takes you to Build Mode.

Build Mode

You can place blocks or spikes.

There's a side panel on the right with buttons.

Click a button, and you're holding that item.

Click the world it places the item.

Play Mode

Both players spawn at the start platform.

Red uses WASD, blue uses arrow keys.

You run, jump, avoid spikes, and try to reach the finish flag.

If you touch the finish, it sends you back to Build Mode.

3. How the Code Is Organized

We split everything into tabs so it's not all in one giant file.

Main.pde

This is the "brain" of the whole game.

It decides which screen to show (start, build, play).

It creates the players.

It handles input like key presses and mouse clicks.

It also stores the lists of blocks and spikes.

BuildMode.pde

This draws everything you see while building:

- the world area
- the side panel
- the buttons
- all the blocks/spikes you placed
- the preview of the item you're holding

It also shows the instructions like "Press P to Play."

PlayMode.pde

This runs the actual gameplay:

- draws the platforms
- draws all blocks and spikes
- updates both players
- checks if someone reached the finish

LevelSaveLoad.pde

This handles saving and loading the level.

Press S saves all block and spike positions to a text file.

Press L loads them back in.

Player.pde

This is the player class.

It handles:

- movement
- gravity
- jumping
- collisions
- spike detection
- drawing the player

Red uses WASD.

Blue uses the arrow keys.

Platform.pde

Just a simple rectangle platform.

Used for the start and finish platforms.

Block.pde

This is the normal block you place in Build Mode.

It's always 50x50.

It has a color and a position.

SpikeBlock.pde

This is the spike trap.

Also 50x50.

Drawn as a red triangle.

If a player touches it, they reset to the start.

StartFinish.pde

This sets up the start and finish platforms and the flags.

It also checks if a player reached the finish.

4. How the Engine Actually Runs all together

1. The start screen shows
2. Click anywhere on the screen to activate Build Mode
3. Place blocks/spikes
4. Press P to activate Play Mode
5. Players move and try to reach the finish
6. If someone reaches the finish line, they get sent back to Build Mode
7. Repeat

The engine keeps everything in lists, so the blocks stay where you put them.

The players use simple physics (gravity, velocity).

Collisions only check vertically to keep things simple.

5. Why We Built It This Way

We wanted something:

- easy to understand
- easy to add new features to
- not too complicated
- good enough to build a real game later

Splitting everything into tabs makes it easier to find stuff.

Using simple shapes (rectangles, triangles) makes testing faster.

Saving and loading help us reuse levels.

6. What We Can Add Later

This engine is still a work in progress, but it’s just the base.

Future ideas:

- more item types
- deleting items
- rotating items
- better movement (dash, wall jump, double jump)
- scoring system
- rounds
- better death effects
- more traps